

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2013

Dresden 500 kV
May 27, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2013. The Interconnection Customer proposes a 1500.0 MW natural gas combined cycle facility interconnecting at the Dresden 500 kV substation in the Ohio area. The request was received on February 20, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 635 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$101,053,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Pt Arena to Cross Plains 230 kV	Pt Arena to W Canton 500 kV	119.2	707	5.9
North Cambridge Substation to Cross Plains 230 kV line	West Canton to Point Arena 500 kV	116.2	847	5.8
Lerdo to Tidd 500 kV	El Paso to Ravenna 500 kV	101.4	2,594	14.0
Dresden 500/230kV Transformer Bank	Dresden Substation to Conesville Sta 500 kV	101.2	976	15.1
Sunnyvale to Conesville Station Substation 230 kV Line	Conesville Station to Dresden 500 kV	100.3	665	10.9
Sunnyvale to Dresden 230 kV	Dresden to Conesville Sta 500 kV Line	100.2	701	10.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Pt Arena to Cross Plains 230 kV	Rebuild line with higher-capacity conductor	\$76,246,000	73% (\$55,578,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cross Plains to North Cambridge 230 kV	Rebuild line with higher-capacity conductor	\$20,571,000	67% (\$13,857,000)
Lerdo to Tidd 500 kV	Reconductor line	\$20,349,000	12% (\$2,419,000)
Dresden 500/230kV Transformer Bank	Replace transformer with higher-rated unit	\$40,786,000	60% (\$24,363,000)
Sunnyvale to Conesville Station 230 kV	Reconductor line	\$11,217,000	33% (\$3,752,000)
Sunnyvale to Dresden 230 kV	Reconductor line	\$3,242,000	33% (\$1,084,000)
	Total allocated to project		\$101,053,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.